

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY

NTA LEVEL 4 - BASIC TECHNICIAN CERTIFICATE
CONTINUOUS ASSESSMENT TEST - I - NOVEMBER-2024
CST 04104 - BASIC PROGRAMMING [CSE/IT]

SEMESTER: I

YEAR: 1

DURATION: 1 Hr 30 Min.

MAX. MARKS: 50

CURATED ANSWERS PROVIDED BY 4LAZIE - SMART IN BRAIN

PART A: SHORT ANSWER QUESTIONS

Q1. List two examples of low level languages.

Answer: Machine Language and Assembly Language.

Q2. Define an array.

Answer: An array is a collection of elements of the same data type stored in contiguous memory locations.

Q3. Write the syntax for do while loop.

Answer:

```
do {  
    // block of code to be executed  
} while (condition);
```

Q4. A self-contained block of code that performs a specific task is known as _____

Answer: Function

Q5. What is a variable?

Answer: A variable is a named storage location in memory used to hold a value that can be changed during the execution of a program.

Q6. Write a c program to print any text.

Answer:

```
#include <stdio.h>  
int main() {  
    printf("Hello, World!");  
    return 0;  
}
```

Q7. Give four types of string functions.

Answer:

- **strlen():** Returns the length of a string.
- **strcpy():** Copies one string to another.
- **strcat():** Concatenates (joins) two strings together.

- **strcmp():** Compares two strings.

PART B: EXPLANATORY QUESTIONS

Q8. Briefly explain any four types of operators.

Answer:

- **Arithmetic Operators:** Used to perform mathematical operations (e.g., +, -, *, /, %).
- **Relational Operators:** Used to compare two values (e.g., ==, !=, >, <).
- **Logical Operators:** Used to combine conditional statements (e.g., &&, ||, !).
- **Assignment Operators:** Used to assign values to variables (e.g., =, +=, -=).

Q9. Write a c program to print even numbers using while loop.

Answer:

```
#include <stdio.h>
int main() {
    int i = 2;
    while(i <= 10) {
        printf("%d ", i);
        i += 2;
    }
    return 0;
}
```

Q10. Write the output on the following program.

```
#include<stdio.h>
int main()
{
    int a=10,b=20;
    printf("results of the number:\n");
    Printf("%d\n",a+b);
    return 0;
}
```

Answer: Assuming the typo 'Printf' (capital P) is corrected to 'printf', the output will be:

```
results of the number:
30
```

Smart Tip: C language is case-sensitive. Using 'Printf' instead of 'printf' in an actual compiler would result in an error.

Q11. With example explain about post increment and pre increment.

Answer:

- **Pre-increment (++a):** The value is incremented first, and then used in the expression.
- **Post-increment (a++):** The value is used in the expression first, and then it is incremented.

Example Code:

```
int a = 5;
int b = ++a; // Pre-increment: a becomes 6, b gets 6
int c = a++; // Post-increment: c gets 6, a becomes 7
```

Q12. Discuss in detail about the types of programming languages.

Answer:

Programming languages are generally divided into two main types:

- **Low-Level Languages:** These are closer to machine hardware. They execute very fast but are hard for humans to read and write. Examples include Machine Language (0s and 1s) and Assembly Language.
- **High-Level Languages:** These are closer to human languages (like English), making them easier to read, write, and maintain. They need to be compiled or interpreted into machine code. Examples include C, Java, and Python.

Q13. Write the syntax for switch statement.

Answer:

```
switch(expression) {
    case value1:
        // code block
        break;
    case value2:
        // code block
        break;
    default:
        // default code block
}
```

PART C: EXPLANATORY AND PROGRAMMING QUESTIONS

Q14. Write c program which involves nested if statement.

Answer:

A nested 'if' statement is an 'if' statement placed inside another 'if' statement.

```
#include <stdio.h>
int main() {
    int num = 10;
    if (num > 0) {
        if (num % 2 == 0) {
            printf("The number is positive and even.\n");
        }
    }
    return 0;
}
```

Q15. Write two separate c programs that involves logical and relational operators.

Answer:

Program 1: Using Relational Operators (>, <, ==)

```
#include <stdio.h>
int main() {
    int a = 10, b = 20;
```

```

    if (a < b) {
        printf("a is less than b\n");
    }
    return 0;
}

```

Program 2: Using Logical Operators (&&, ||)

```

#include <stdio.h>
int main() {
    int age = 20;
    if (age >= 18 && age < 30) {
        printf("You are a young adult.\n");
    }
    return 0;
}

```

Q16. Write a c program that involves conditional operator.

Answer:

The conditional (ternary) operator `?:` is a shorthand for an if-else statement. Syntax: condition ? value_if_true : value_if_false;

```

#include <stdio.h>
int main() {
    int a = 10, b = 20;
    int max = (a > b) ? a : b;
    printf("The maximum number is: %d\n", max);
    return 0;
}

```

Q17. i) Explain in detail about arrays.

Answer:

An array is a fundamental data structure in C that stores a fixed-size sequential collection of elements of the same data type.

- **Declaration:** data_type array_name[array_size]; (e.g., int numbers[5];)
- **Access:** Array elements are accessed using indices, starting from 0. So the first element is numbers[0], and the last is numbers[4].
- **Advantages:** It makes it easy to store and manipulate a large amount of related data using a single variable name.
- **Disadvantages:** The size of the array is fixed at compile time and cannot be changed dynamically.

Q17. ii) Write a c program that displays number of days using switch statement.

Answer:

```

#include <stdio.h>
int main() {
    int day_number = 2;
    switch(day_number) {
        case 1: printf("Monday\n"); break;
        case 2: printf("Tuesday\n"); break;
        case 3: printf("Wednesday\n"); break;
        case 4: printf("Thursday\n"); break;
    }
}

```

```
case 5: printf("Friday\n"); break;
case 6: printf("Saturday\n"); break;
case 7: printf("Sunday\n"); break;
default: printf("Invalid day number\n");
}
return 0;
}
```